GENESIS: Generative Emulation of Cosmological Multifield Maps with Optimal-Transport Flow Matching

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Abstract

Diffusion models have reshaped generative modelling across many domains and have also found wide application in cosmology, including field reconstruction, emulation, and parameter inference. However, diffusion-based approaches typically require many sampling steps and provide only variational lower bounds on the likelihood. Flow matching has recently emerged as an alternative generative framework with stable training, efficient sampling, and tractable likelihood evaluation, but its use in cosmology remains relatively limited. To explore flow matching as a cosmological emulator with efficient sampling and likelihood access, we introduce GENESIS, a conditional continuous normalizing flow trained with optimal-transport flow matching on the IllustrisTNG maps of the CAMELS Multifield Dataset. Given $(\Omega_\mathrm{m}, \sigma_8)$ and the CAMELS supernova- and AGN-feedback parameters, GENESIS generates $25\,h^{-1}\,\mathrm{Mpc}$ two-dimensional maps of dark matter density, gas density, and gas temperature, while jointly modelling their shared structure and baryonic responses. The generated samples reproduce the main one- and two-point statistics of CAMELS while preserving the leading correlations among the three fields, with each multifield realization generated in a few seconds in our implementation. As an initial proof of concept for likelihood-based inference with this flow-matching emulator, we perform a two-dimensional Ω_m–σ_8 relative-likelihood scan for one CAMELS map using the GENESIS continuous-normalizing-flow likelihood, with the remaining parameters fixed. The likelihood surface peaks at the true parameter values, showing that the learned flow assigns informative field-level likelihoods that can be used as a basis for future cosmological parameter inference.

Keywords: Cosmological fields — Cosmological parameters — Generative models — Parameter inference — Neural networks

1 Introduction

In modern cosmology, machine learning has emerged as an indispensable tool across a broad range of problems, including cosmological parameter inference [1, 2], field-level analyses and baryonic field mapping [3, 4], and generative modelling and emulation of cosmological fields [5–7]. As in many other machine-learning domains, progress in these problems depends on access to large and diverse data sets. In cosmology, however, such data sets are often provided by cosmological hydrodynamic simulations that model complex baryonic physics, making them computationally expensive to produce. The Cosmology and Astrophysics with Machine Learning Simulations [CAMELS; 8] project was conceived to address this challenge by providing thousands of simulations in which both cosmological and astrophysical parameters are varied in a controlled fashion. As part of the CAMELS project, the CAMELS Multifield Dataset [CMD; 9] releases these simulations as 2D maps and 3D grids of multiple physical fields, including dark matter density, gas density, and gas temperature, together with their associated simulation parameters. Since these fields are co-registered and share the same underlying CAMELS parameters, CMD provides a natural testbed for studying not only individual field statistics but also correlations among physically related observables.

Meanwhile, in the machine-learning domain, deep generative models have made substantial progress over the past decade. This progress can be viewed as a sequence of advances in which successive model families addressed key limitations of earlier approaches while introducing new trade-offs. The variational autoencoder [VAE; 10] provided an early foundation by learning structured latent representations of data, although it often produced overly smooth samples. Generative adversarial networks [GANs; 11] shifted the focus toward perceptual sample quality by training a generator and a discriminator in an adversarial game, yielding sharper samples but remaining vulnerable to mode collapse. Normalizing flows [12] provided exact likelihood evaluation through invertible mappings between a simple prior and the data distribution, but their architectural constraints often limited their expressiveness. More recently, diffusion models [13, 14] have reshaped the generative modelling landscape by providing a stable and scalable approach to learning complex data distributions through the reversal of a gradual noising process. However, diffusion models are subject to two key limitations: their training objective is tied to an evidence lower bound of the likelihood (ELBO), making exact likelihood evaluation intractable, and sampling typically requires many steps. Flow matching [15] has recently emerged as an alternative that mitigates both limitations. It trains a neural network to approximate the time-dependent velocity field that transports a simple prior distribution into the data distribution, modelling the evolution of the probability path over time. Through its connection to continuous normalizing flows [16], this formulation supports likelihood evaluation via the change-of-variables formula. Furthermore, when combined with optimal-transport couplings [15], flow matching induces straighter probability paths between the two distributions, reducing the number of steps required for sampling.

This methodological progression has also shaped the use of generative models in cosmology. Early approaches applied GANs and normalizing flows to cosmological multifield emulation, HI-map generation, and likelihood-based inference [17–20]. Diffusion models were subsequently adopted for dark-matter reconstruction from biased tracers [21, 22], HI-map emulation [23], and dark matter density-field generation and parameter inference [24, 25]. More recent work has explored latent-diffusion upscaling of cosmological fields [26] and stochastic-interpolant baryonic field mapping [27]. Flow matching has also begun to enter cosmology, for example in Kannan et al. [28], which uses flow matching for scale-aware representation learning of field-level cold dark matter simulations. However, direct conditional generation and emulation of multiple co-registered cosmological fields remains largely unexplored.

This motivates the central objective of this work: to apply optimal-transport flow matching (OT-FM) to direct conditional cosmological field generation and emulation, thereby exploiting key advantages of OT-FM that remain largely unused in cosmological field modelling, including efficient sampling through straighter transport paths and likelihood evaluation within the continuous normalizing flow (CNF) framework [16]. In addition to statistical fidelity, the practical sampling cost of the trained model is quantified: in the current implementation, generating 200 independent multifield map triplets at a fixed conditioning vector takes roughly 20 minutes on a single NVIDIA A100 GPU. Beyond the choice of generative framework, existing generative models built on CAMELS/CMD data have typically used only part of the available conditioning information, focusing primarily on the two cosmological parameters ($\Omega_{\rm m}$, σ_8) [17, 18, 29, 23–25]. The four astrophysical feedback parameters varied in CAMELS ($A_{\rm SN1}$, $A_{\rm SN2}$, $A_{\rm AGN1}$, $A_{\rm AGN2}$) have therefore generally been fixed, ignored, or marginalized over, despite their strong influence on baryonic fields. At the same time, most existing generative models have targeted individual fields in isolation [19, 29, 23, 24]. Together, these limitations point to the need for a fully conditioned multifield generative emulator that can model correlations among physically coupled fields.

This work introduces GENESIS (**GEN**erative **E**mulator for **S**imulation and parameter **I**nference from cosmological field**s**), a conditional continuous normalizing flow trained with OT-FM for multifield cosmological emulation and field-level likelihood-based parameter inference. The main contributions are as follows.

1. GENESIS appears to be the first generative model to apply OT-FM to direct conditional generation of multifield cosmological maps. While GAN-based models have previously generated multiple cosmological fields [17, 18], no prior work has produced M_{cdm}, M_{gas}, and T jointly.
2. GENESIS conditions on the full six-dimensional CAMELS parameter vector associated with each CMD map, namely the two cosmological parameters (Ω_{m}, σ_8) and the four astrophysical feedback parameters (A_{SN1}, A_{SN2}, A_{AGN1}, A_{AGN2}). This conditioning enables the model to capture the distinct responses of dark matter density, gas density, and gas temperature to cosmological and feedback variations across the CAMELS design space. No previous generative model appears to have produced multifield maps conditioned on this full parameter vector.
3. The inference potential of GENESIS is illustrated without additional training. Through the CNF change-of-variables formula, the model assigns likelihoods to input multifield maps under candidate conditioning vectors. As a proof of concept, one held-out CMD multifield map $\mathbf{x}_1$ is fixed and the $(\Omega_{\mathrm{m}}, \sigma_8)$ plane is scanned, with the four astrophysical feedback parameters held at their true values. Within the evaluated grid, the maximum-likelihood point coincides with the true CAMELS parameter point, and the Wilks-theorem 1σ contour encloses neighbouring grid points. This is therefore framed as evidence that the likelihood surface is centred on the truth at the scan resolution, rather than as a unique parameter resolution within stochastic noise.

The remainder of this paper is organized as follows. Section 2 describes the CAMELS simulations, the CMD field data, and the field choices. Section 3 presents the OT-FM framework and GENESIS architecture. Sections 4 and 5 evaluate GENESIS on a test split of the CAMELS/CMD Latin-hypercube set and on an independent constant-parameter set, respectively. Section 6 presents a likelihood scan as a proof of concept, demonstrating the potential of GENESIS for field-level parameter inference. Section 7 discusses the progress and limitations of GENESIS in the context of existing work, and outlines potential future directions for improving the model and extending its applications. Finally, Section 8 summarizes the findings and concludes the paper.

2 CAMELS Multifield Dataset

This section describes the publicly released 2D maps from the CAMELS Multifield Dataset (CMD) used by GENESIS (Section 2.1), the three target channels (M_{cdm}, M_{gas}, and T) and the IllustrisTNG data splits adopted in this work (Section 2.2), and the six conditioning parameters and their field-level sensitivities (Section 2.3).

2.1 IllustrisTNG 2D maps

The analysis is based on the IllustrisTNG 2D maps from the CMD [9]. These maps are derived from hydrodynamic runs that adopt the baryonic model of Weinberger et al. [30] and Pillepich et al. [31]. These simulations evolve a periodic box of side $25\,h^{-1}$ Mpc with 256^3 dark matter particles and an equal number of initial gas cells from $z = 127$ to $z = 0$. Within the public IllustrisTNG map release, the LH set is used for training, validation, and testing, and the CV set is used to assess statistical consistency across independent realizations at fixed parameter values. For each simulation and target channel, the release provides 15 projected two-dimensional maps, obtained from five non-overlapping slices along each of the three Cartesian axes.

LH set. One thousand simulations sampling the six-dimensional CAMELS parameter space with a Latin-hypercube design (see Table 1 for parameter ranges). This parameter-varying set is used for training, validation, and testing, and is partitioned into 800 training, 100 validation, and 100 test simulations.

CV set. Twenty-seven simulations at the fiducial parameter vector $(\Omega_{\mathrm{m}}, \sigma_8, A_{\mathrm{SN1}}, A_{\mathrm{SN2}}, A_{\mathrm{AGN1}}, A_{\mathrm{AGN2}}) = (0.30, 0.80, 1, 1, 1, 1)$, differing only in their initial conditions. This set is used to test whether generated samples are statistically consistent with independent simulations at the same parameter point.

2.2 Field definitions

Three CMD fields are selected as the joint emulation target of GENESIS: projected dark matter surface mass density (M_{cdm}), projected gas surface mass density (M_{gas}), and projected mass-weighted gas

Parameter	LH range	Physical meaning
Ω_{m}	$[0.1,\ 0.5]$	Total matter density fraction
σ_8	$[0.6,\ 1.0]$	Linear fluctuation amplitude
A_{SN1}	$[0.25,\ 4.0]$	SN wind energy per SFR
A_{SN2}	$[0.5,\ 2.0]$	SN wind speed
A_{AGN1}	$[0.25,\ 4.0]$	AGN energy per BH accretion rate
A_{AGN2}	$[0.5,\ 2.0]$	AGN kinetic ejection speed

Table 1: CAMELS IllustrisTNG conditioning parameters, their Latin-hypercube ranges, and physical interpretations [8, 9].

temperature (T). Each map comprises 256×256 pixels covering a field of view of $(25\,h^{-1}\,\mathrm{Mpc})^2$, extracted from the $z = 0$ snapshot, and is constructed by projecting particle properties onto a regular grid using an adaptive smoothing kernel [9].

Together, these three fields encode the dominant matter components and the thermodynamic state of the gas. Their cross-correlations provide sensitive diagnostics of baryonic feedback, motivating the joint emulation approach of GENESIS.

2.3 Conditioning parameters and field-level sensitivity

GENESIS is conditioned on the full six-dimensional CAMELS parameter vector

$$\boldsymbol{\theta} = (\Omega_{\mathrm{m}},\, \sigma_8,\, A_{\mathrm{SN1}},\, A_{\mathrm{SN2}},\, A_{\mathrm{AGN1}},\, A_{\mathrm{AGN2}}), \tag{1}$$

which characterizes each IllustrisTNG simulation in CMD [8, 9]. Table 1 summarizes the Latin-hypercube ranges and physical meaning of each parameter.

The two *cosmological parameters* govern the overall amplitude and shape of the matter power spectrum. Ω_{m} sets the total matter density of the Universe, while σ_8 defines the root-mean-square amplitude of linear density fluctuations smoothed on $8\,h^{-1}$ Mpc scales. Both parameters primarily govern dark matter clustering, with the baryonic fields responding through the evolving gravitational potential. The four *astrophysical feedback parameters* encode the subgrid prescriptions of the IllustrisTNG model [30, 31]. A_{SN1} and A_{SN2} control two properties of supernova-driven galactic winds: the energy emitted per unit star-formation rate and the wind speed, respectively. A_{AGN1} and A_{AGN2} represent the energy released per unit black hole accretion rate and the ejection speed of the kinetic mode of black hole feedback, respectively. These feedback parameters have a much weaker direct effect on the collisionless dark matter field, but strongly modulate the baryonic fields [8]. Supernova feedback expels gas from low-mass haloes, suppressing the gas density power spectrum at small scales, while AGN feedback operates predominantly in massive haloes and heats the surrounding gas, dominating the temperature power spectrum at cluster scales.

3 Methodology

This section describes the methodology of GENESIS. Section 3.1 introduces the optimal-transport flow matching (OT-FM), Section 3.2 presents the U-Net architecture and its physically motivated design choices, and Section 3.3 summarizes the training procedure. Full optimisation settings are given in Appendix A.

3.1 Optimal-transport flow matching

Flow matching, introduced by Lipman et al. [15], is a class of generative models that learns a time-dependent flow ψ_t transporting samples from a simple prior distribution p_0 to the data distribution p_1. The flow is defined through the ordinary differential equation (ODE)

$$\frac{\mathrm{d}\mathbf{x}_t}{\mathrm{d}t} = u_t(\mathbf{x}_t), \qquad \mathbf{x}_0 \sim p_0, \tag{2}$$

where u_t is the time-dependent velocity field and $\mathbf{x}_t = \psi_t(\mathbf{x}_0)$. The ideal flow-matching objective regresses a neural velocity field u_t^{ϕ} onto the target vector field associated with a probability path

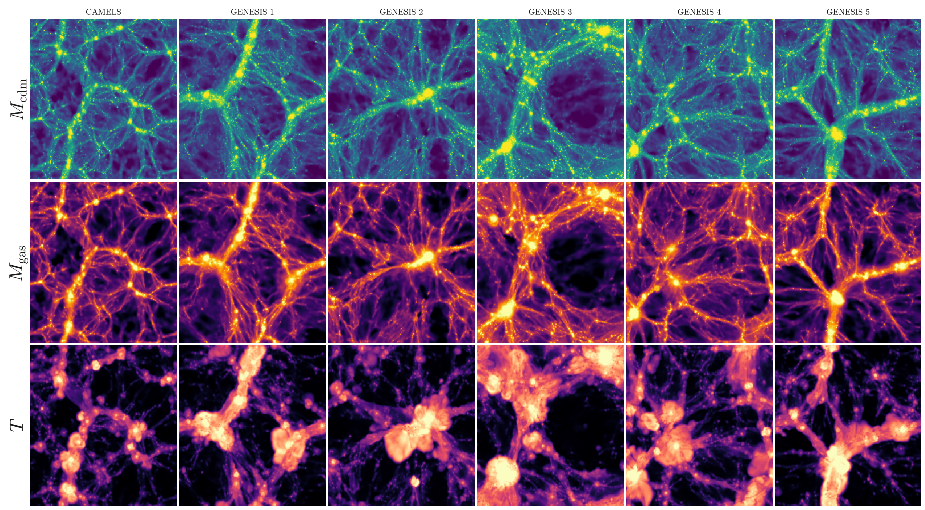

Figure 1: Comparison between CAMELS LH maps and GENESIS-generated maps. The first column shows a CAMELS map triplet, while the remaining columns show independent GENESIS samples generated at the same conditioning vector, $(\Omega_{\mathrm{m}}, \sigma_8, A_{\mathrm{SN1}}, A_{\mathrm{SN2}}, A_{\mathrm{AGN1}}, A_{\mathrm{AGN2}}) = (0.3542, 0.8706, 1.3398, 0.8839, 1.0785, 1.0063)$. Rows show M_{cdm}, M_{gas}, and T.

$p_t(\mathbf{x} \mid \boldsymbol{\theta})$ connecting p_0 to the conditional data distribution $p_1(\mathbf{x} \mid \boldsymbol{\theta})$:

$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}_{t\sim\mathcal{U}(0,1),\, \mathbf{x}\sim p_t(\mathbf{x}|\boldsymbol{\theta})} \left\| u_t^{\phi}(\mathbf{x} \mid \boldsymbol{\theta}) - u_t(\mathbf{x} \mid \boldsymbol{\theta}) \right\|^2, \tag{3}$$

where ϕ denotes the network parameters. However, the target vector field $u_t(\mathbf{x} \mid \boldsymbol{\theta})$ is generally intractable. The conditional flow-matching (CFM) formulation of Lipman et al. [15] is therefore used. Here, "conditional" refers to auxiliary probability paths conditioned on data samples $\mathbf{x}_1$, and is distinct from the CAMELS conditioning vector $\boldsymbol{\theta}$. CFM defines the tractable objective

$$\mathcal{L}_{\mathrm{CFM}} = \mathbb{E}_{t\sim\mathcal{U}(0,1),\, \mathbf{x}_1\sim q(\mathbf{x}_1|\boldsymbol{\theta}),\, \mathbf{x}\sim p_t(\mathbf{x}|\mathbf{x}_1)} \left\| u_t^{\phi}(\mathbf{x} \mid \boldsymbol{\theta}) - u_t(\mathbf{x} \mid \mathbf{x}_1) \right\|^2. \tag{4}$$

A key result of Lipman et al. [15] is that this conditional objective has the same gradient with respect to the network parameters as the ideal flow-matching objective,

$$\nabla_{\phi}\mathcal{L}_{\mathrm{FM}}(\phi) = \nabla_{\phi}\mathcal{L}_{\mathrm{CFM}}(\phi). \tag{5}$$

Thus, training can proceed by specifying the conditional probability path $p_t(\mathbf{x} \mid \mathbf{x}_1)$ and its associated velocity field, without explicitly evaluating $u_t(\mathbf{x})$.

In this work, the optimal-transport version of CFM, hereafter OT-FM, is adopted. For each conditioning vector $\boldsymbol{\theta}$, target samples $\mathbf{x}_1 \sim q(\mathbf{x}_1 \mid \boldsymbol{\theta})$ and source samples $\mathbf{x}_0 \sim p_0$ are drawn. Following Lipman et al. [15], the conditional probability path is taken to be a Gaussian path centred on the linear interpolant,

$$p_t(\mathbf{x} \mid \mathbf{x}_1) = \mathcal{N}\big(\mathbf{x}; t\,\mathbf{x}_1,\, (1-t)^2\mathbf{I}\big), \quad t \in [0, 1]. \tag{6}$$

Using the reparameterization trick [10], this path can be written as

$$\mathbf{x}_t = (1-t)\mathbf{x}_0 + t\mathbf{x}_1, \quad \mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{7}$$

with the associated conditional vector field

$$\frac{\mathrm{d}\mathbf{x}_t}{\mathrm{d}t} = \mathbf{x}_1 - \mathbf{x}_0. \tag{8}$$

Training therefore minimizes

$$\mathcal{L}_{\mathrm{CFM}} = \mathbb{E}_{t\sim\mathcal{U}(0,1),\, \mathbf{x}_0\sim p_0,\, \mathbf{x}_1\sim q(\mathbf{x}_1|\boldsymbol{\theta})} \left\| u_t^{\phi}(\mathbf{x}_t \mid \boldsymbol{\theta}) - (\mathbf{x}_1 - \mathbf{x}_0) \right\|^2. \tag{9}$$

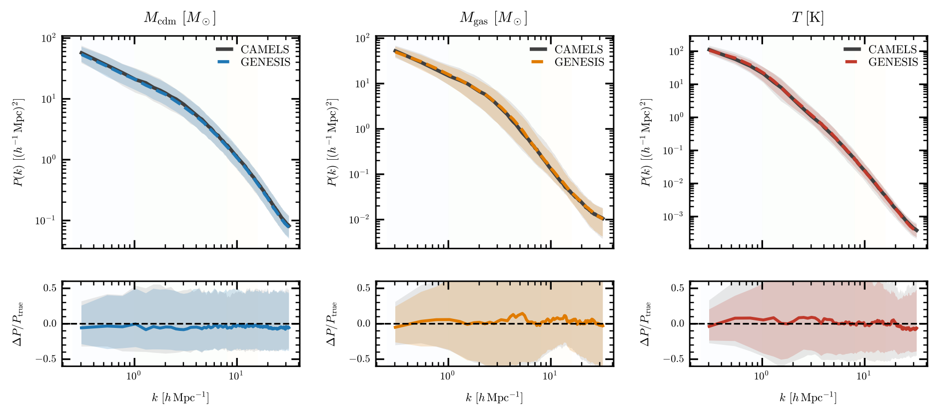

Figure 2: Auto-power spectra and fractional residuals on the LH test set. Top panels show the LH-aggregated CAMELS and GENESIS auto-power spectra for $M_{\rm cdm}$, $M_{\rm gas}$, and T after aggregating over the 100 held-out LH conditioning vectors. Bottom panels show $\epsilon_P(k)$, the fractional difference between the LH-aggregated mean spectra defined in equation (12).

Sampling. After training, to generate a sample at a specified conditioning vector $\boldsymbol{\theta}$, a source sample $\mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is drawn and equation (2) is integrated from $t = 0$ to $t = 1$ using the Dormand–Prince (`dopri5`) adaptive Runge–Kutta solver with relative and absolute tolerances of 10^{-5}. The terminal state $\mathbf{x}_1$ is interpreted as a sample from the learned conditional data distribution. For the CV sampling runs used below, generating 200 independent multifield map triplets at a fixed conditioning vector takes roughly 20 minutes on a single NVIDIA A100 GPU, corresponding to about 6 seconds per triplet in the current implementation.

Classifier-free guidance. Classifier-free guidance [CFG; 32] is commonly used in conditional generation by training a model on both conditional and unconditional inputs. In the flow-matching setting, the same idea can be applied at the level of the learned velocity field: a guided velocity is formed by extrapolating the conditional velocity away from the unconditional one,

$$\hat{u}_t = u_t^\phi(\mathbf{x}_t \mid \boldsymbol{\theta}_\varnothing) + w\left[u_t^\phi(\mathbf{x}_t \mid \boldsymbol{\theta}) - u_t^\phi(\mathbf{x}_t \mid \boldsymbol{\theta}_\varnothing)\right], \tag{10}$$

where w is the guidance weight and $\boldsymbol{\theta}_\varnothing$ denotes the null token. Values $w > 1$ strengthen the influence of the conditioning vector by amplifying the difference between the conditional and unconditional velocity predictions. During training, the CAMELS conditioning vector $\boldsymbol{\theta}$ is randomly replaced with $\boldsymbol{\theta}_\varnothing$ with probability $p_{\rm cfg} = 0.1$, exposing the network to both conditional and unconditional velocity prediction tasks. In this work, $w = 1$ is used at inference, which recovers the standard conditional velocity $\hat{u}_t = u_t^\phi(\mathbf{x}_t \mid \boldsymbol{\theta})$ without extrapolation. Thus, the null-token branch is used primarily as a training-time conditioning dropout mechanism rather than as an inference-time guidance boost. Empirically, retaining this branch improved both the validation loss and the statistical quality of generated samples, suggesting that exposure to unconditional inputs regularizes the learned velocity field.

3.2 U-Net architecture

The velocity field u_t^ϕ is parameterized with a U-Net architecture equipped with skip connections [33]. The network takes as input the interpolated field $\mathbf{x}_t \in \mathbb{R}^{3\times256\times256}$, the flow time t, and the conditioning vector $\boldsymbol{\theta} \in \mathbb{R}^6$. The flow time t is mapped to a sinusoidal embedding, and the conditioning vector $\boldsymbol{\theta}$ is mapped through a four-layer MLP, each to 512 dimensions. These embeddings are concatenated and projected through an MLP $(1024 \to 1024 \to 512)$ to form a shared time–condition embedding $\mathbf{e}_{\rm joint}$.

This backbone is augmented with three physically motivated design choices. **Circular convolutions.** The projected maps inherit periodic boundary conditions from the CAMELS simulation volume [9]. Accordingly, circular padding is used in all convolutional layers, reducing boundary artefacts that would be introduced by zero or reflective padding. **Stage-wise conditioning.** At each resolution level, the shared time–condition embedding $\mathbf{e}_{\mathrm{joint}}$ is augmented with a stage-specific projection of the conditioning embedding,

$$\boldsymbol{\ell}_i = \mathbf{e}_{\mathrm{joint}} + W_i(\mathbf{c}_{\mathrm{emb}}), \tag{11}$$

where $\mathbf{c}_{\mathrm{emb}}$ denotes the conditioning embedding and W_i is a learned linear adapter for stage i. The resulting level embedding $\boldsymbol{\ell}_i$ is then used to modulate the GroupNorm [34] output at every residual block via learned affine scale and shift parameters [FiLM; 35]. This allows each resolution level to interpret the conditioning information differently: coarse stages govern large-scale morphology and power-spectrum amplitude, while finer stages modulate small-scale texture and the spatial extent of feedback-driven features. **Scale-limited self-attention.** Self-attention [36] is applied at 64×64 resolution and below, including the bottleneck and corresponding decoder stages. By computing pairwise similarities across all spatial positions, it captures the long-range correlations and inter-field dependencies that local convolutions cannot reach. The final architecture has four encoder stages with channel widths $[128, 256, 384, 512]$, two residual blocks per encoder stage, strided circular convolutions for downsampling, a residual–attention–residual bottleneck, and a symmetric decoder with skip concatenations. The network maps three input channels to three output velocity channels and contains approximately 93.0 million trainable parameters.

3.3 Training

GENESIS is trained with the OT-FM objective using AdamW optimisation [37], D4 data augmentation, and the classifier-free conditioning dropout described above. The final checkpoint is obtained through an initial training stage followed by two fine-tuning stages, with the last stage disabling conditioning dropout. All results reported below use this final checkpoint. Full optimisation settings are provided in Appendix A.

4 LH evaluation

This section evaluates GENESIS on the held-out CAMELS/CMD Latin-hypercube (LH) test set. The LH test set contains 100 held-out simulations, each with 15 projected multifield map triplets; for each conditioning vector, 30 GENESIS samples are generated. Section 4.1 presents representative generated map triplets, Section 4.2 compares the auto-power spectra of the three target fields, Section 4.3 evaluates one-point field distributions and mean amplitudes, Section 4.4 examines cross-power spectra and inter-field coherence, and Section 4.5 assesses the conditional parameter response and ensemble spread.

4.1 Field morphology

A visual comparison of GENESIS maps from the LH test set is first presented. Since GENESIS is a stochastic generator, the generated maps are not expected to reconstruct the same realization as the CAMELS maps. Instead, this comparison checks whether samples generated at the same conditioning vector exhibit similar field morphology and cross-field structure.

Figure 1 shows that GENESIS produces visually plausible samples with filamentary dark-matter structure and gas distributions that trace the cosmic web. The generated maps differ from the CAMELS map at the realization level, as expected for a stochastic generator without initial-condition conditioning. The dark-matter and gas-density channels show comparable large-scale morphology, while the temperature channel exhibits softer small-scale features in the generated samples, a qualitative impression that is revisited quantitatively in Section 4.5.

4.2 Auto-power spectra

The auto-power spectrum $P_a(k)$ is next compared for each target field $a \in \{M_{\mathrm{cdm}}, M_{\mathrm{gas}}, T\}$. Here $P_a(k)$ is computed from the two-dimensional overdensity field of each map and azimuthally averaged in Fourier space. This statistic measures the scale-dependent fluctuation amplitude of each field, providing a direct test of whether the generated maps reproduce the clustering strength of dark matter, gas, and temperature structures.

In this subsection, statistics are aggregated over the 100 held-out LH conditioning vectors. Thus, the comparison tests the marginal distribution of power spectra across the LH test set. The parameter-

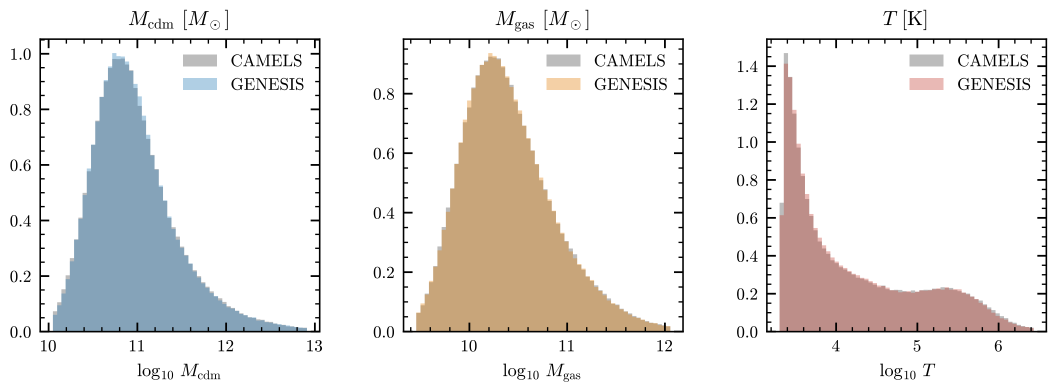

Figure 3: One-point pixel distributions on the LH test set in $\log_{10}$ field space. Grey histograms show CAMELS and coloured histograms show GENESIS.

dependent response is examined separately in Section 4.5. For each field, the fractional difference between the LH-aggregated GENESIS and CAMELS mean auto-power spectra is computed as

$$\epsilon_P(k) = \frac{|\langle P_a^{\rm gen}(k;\theta)\rangle_\theta - \langle P_a^{\rm true}(k;\theta)\rangle_\theta|}{\langle P_a^{\rm true}(k;\theta)\rangle_\theta}, \tag{12}$$

where the average is taken over the held-out LH conditioning vectors. $\epsilon_P(k)$ is then averaged within three reporting bands: low-k ($k < 1$), mid-k ($1 \leq k < 8$), and high-k ($8 \leq k < k_{\rm art}$), with k in units of $h\,{\rm Mpc}^{-1}$ and $k_{\rm art} \simeq 16$ denoting the upper edge used before the reference-only artefact regime.

Figure 2 shows that the generated auto-power spectra closely follow the LH-aggregated CAMELS reference spectra for all three fields. The dark-matter channel provides the cleanest gravitational baseline, with particularly small residuals at high k. The baryonic fields, $M_{\rm gas}$ and T, are more sensitive to supernova and AGN feedback, but their generated spectra remain close to the CAMELS reference over the reported scale bands.

Channel	$\bar{\epsilon}_P^{\rm low}$	$\bar{\epsilon}_P^{\rm mid}$	$\bar{\epsilon}_P^{\rm high}$	KS	JSD	R^2
$M_{\rm cdm}$	3.9%	2.1%	0.5%	0.0336	0.00262	0.842
$M_{\rm gas}$	4.2%	5.3%	4.7%	0.0364	0.00293	0.882
T	2.7%	1.9%	3.6%	0.0464	0.00454	0.833

Table 2. Quantitative evaluation on the LH test set ($N_{\rm sim} = 100$, $n_{\rm gen} = 30$ per sim). Auto-$P(k)$ errors $\bar{\epsilon}_P$ are band-averaged fractional differences between the LH-aggregated CAMELS and GENESIS mean auto-power spectra, following equation (12). KS and JSD measure pixel-value distribution fidelity. R^2 is the overall conditional response quality.

The band-averaged errors in Table 2 summarize the fractional difference between the LH-aggregated CAMELS and GENESIS mean spectra. For $M_{\rm cdm}$, the residuals are 3.9, 2.1, and 0.5 per cent in the low-, mid-, and high-k bands, respectively. For $M_{\rm gas}$, the corresponding residuals are 4.2, 5.3, and 4.7 per cent, while for T they are 2.7, 1.9, and 3.6 per cent. These values therefore measure agreement with the marginal spectral distribution over the held-out LH set. The R^2 column in Table 2 instead summarizes the conditional response in log-power space and is discussed in Section 4.5. Overall, the auto-power spectra indicate that GENESIS reproduces the marginal distribution of scale-dependent fluctuation amplitudes across the held-out LH parameter space.

4.3 One-point field distributions and mean amplitudes

The auto-power spectra above are computed from overdensity fields and primarily test relative spatial fluctuations. One-point field statistics are also examined in physical field space. These diagnostics test whether GENESIS reproduces the absolute amplitude range of each field, including dense matter and gas regions as well as the high-temperature tail of shock- and feedback-heated gas.

Figure 3 shows that the pixel distributions of GENESIS and CAMELS are very similar for all three target fields. The $M_{\rm cdm}$ and $M_{\rm gas}$ distributions nearly overlap, while T is more challenging due

Metric	$M_{\rm cdm}$	$M_{\rm gas}$	T
KS stat D	0.0292 $[0.0135,\ 0.0574]$ (82%)	0.0328 $[0.0149,\ 0.0568]$ (78%)	0.0408 $[0.0214,\ 0.0734]$ (58%)
ε_μ	0.0018 $[0.0006,\ 0.0038]$ (100%)	0.0022 $[0.0007,\ 0.0047]$ (100%)	0.0097 $[0.0034,\ 0.0231]$ (100%)
ε_σ	0.0171 $[0.0044,\ 0.0363]$ (100%)	0.0161 $[0.0044,\ 0.0441]$ (99%)	0.0247 $[0.0070,\ 0.0570]$ (98%)
JSD	0.0020 $[0.0014,\ 0.0039]$	0.0022 $[0.0014,\ 0.0041]$	0.0038 $[0.0022,\ 0.0063]$

Table 3: LH test-set pixel-value distribution statistics ($N_{\rm sim} = 100$, $n_{\rm gen} = 30$ per sim). Each cell reports the median and 68% interval [$\rm p_{16}$, $\rm p_{84}$] over the 100 simulations. Pass% indicates the fraction of simulations below the acceptance threshold (KS ≤ 0.05; $\varepsilon_\mu \leq 0.05$; $\varepsilon_\sigma \leq 0.10$). JSD has no threshold and is reported for reference.

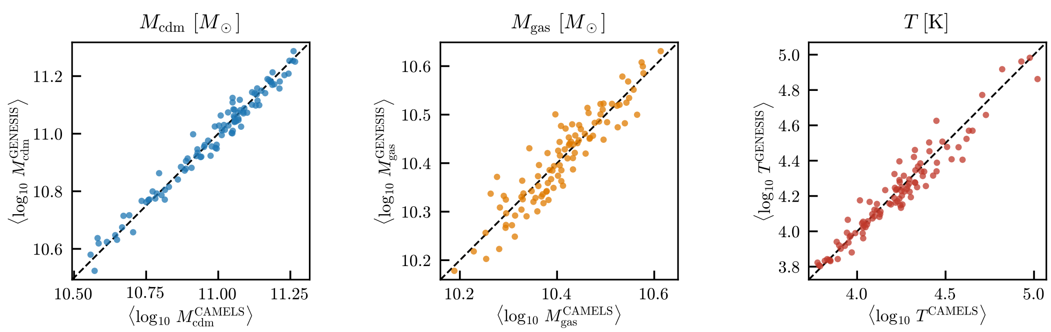

Figure 4: Per-simulation mean field values on the LH test set. Each point corresponds to one held-out conditioning vector. The dashed line marks the one-to-one relation between CAMELS and GENESIS.

to the combination of cold diffuse gas with shock- and feedback-heated material. Nevertheless, the GENESIS temperature distribution reproduces the CAMELS distribution, including its extended high-temperature tail and two modes.

The pixel distributions in $\log_{10}$ field space are then compared using the Kolmogorov–Smirnov statistic $D_{\rm KS}$, the Jensen–Shannon divergence (JSD), and the relative errors in the mean and standard deviation,

$$\varepsilon_\mu = \frac{|\mu_{\rm gen} - \mu_{\rm true}|}{|\mu_{\rm true}|}, \qquad \varepsilon_\sigma = \frac{|\sigma_{\rm gen} - \sigma_{\rm true}|}{\sigma_{\rm true}}. \tag{13}$$

Table 3 summarizes these one-point statistics across the 100 LH test simulations. The median KS statistics are 0.0292, 0.0328, and 0.0408 for $M_{\rm cdm}$, $M_{\rm gas}$, and T, respectively, while the corresponding JSD values are 0.0020, 0.0022, and 0.0038. The mean field amplitudes are also accurately reproduced: the median relative mean errors are 0.0018, 0.0022, and 0.0097 for $M_{\rm cdm}$, $M_{\rm gas}$, and T. The relative errors in the pixel standard deviation remain small as well, with median values of 0.0171, 0.0161, and 0.0247. Thus, GENESIS matches not only the spatial fluctuations measured by $P(k)$, but also the absolute amplitude scale and width of the one-point field distributions.

Figure 4 compares the mean field value for each held-out conditioning vector. The points lie close to the one-to-one relation for all three channels, showing that GENESIS tracks the parameter-dependent mean amplitude of the CAMELS maps. The agreement is tightest for $M_{\rm cdm}$, while $M_{\rm gas}$ and T show larger scatter, consistent with their stronger sensitivity to astrophysical feedback.

4.4 Cross-power spectra and inter-field coherence

The preceding diagnostics test each field individually. The mutual consistency of the three generated fields is then assessed. This is essential for a multifield emulator: matching the auto-power spectra and one-point distributions of $M_{\rm cdm}$, $M_{\rm gas}$, and T separately does not guarantee that their spatial correlations are correct.

The joint structure is quantified using the cross-power spectrum

$$P_{ab}(k) = \mathrm{Re}\left[\left\langle \tilde{\delta}_a(\mathbf{k})\tilde{\delta}_b^*(\mathbf{k})\right\rangle_{|\mathbf{k}|=k}\right], \tag{14}$$

and the inter-field coherence

$$r_{ab}(k) = \frac{P_{ab}(k)}{\sqrt{P_{aa}(k)P_{bb}(k)}}. \tag{15}$$

The cross-power measures the scale-dependent amplitude of shared structure between two fields, while the coherence removes the individual field amplitudes and isolates their normalized spatial correlation.

Figure 5 shows that GENESIS reproduces the main cross-field correlations of the CAMELS maps. The M_{cdm}–M_{gas} pair is matched most closely, as expected from the strong coupling between dark matter and gas density through the underlying gravitational structure. The temperature-related pairs are more challenging, because T is affected not only by the density field but also by shock heating and feedback processes. Nevertheless, the generated cross-power spectra and coherence curves follow the CAMELS trends across the reported scales.

This agreement indicates that GENESIS does not merely generate three plausible fields independently. Instead, the generated map triplets preserve the correlated spatial structure among dark matter density, gas density, and gas temperature. The largest visible differences occur for temperature-related pairs, especially on smaller scales where baryonic feedback introduces additional stochasticity.

4.5 Conditional response and ensemble spread

The diagnostics above show that GENESIS reproduces the marginal field statistics and the cross-field correlations of the LH test set. Whether the model also tracks how these statistics vary across the six-dimensional CAMELS conditioning space is then examined. For the auto-power spectrum, this response is quantified using

$$R^2(k) = 1 - \frac{\mathrm{MSE}_\theta\left[\log P_a^{\mathrm{gen}}(k;\theta) - \log P_a^{\mathrm{true}}(k;\theta)\right]}{\mathrm{Var}_\theta\left[\log P_a^{\mathrm{true}}(k;\theta)\right]}, \tag{16}$$

where the variance is taken over the 100 held-out LH simulations. A value of $R^2 = 1$ indicates perfect tracking of the parameter-dependent log-power response, while $R^2 = 0$ corresponds to the performance of a parameter-independent mean prediction.

The normalized within-condition spread is also examined,

$$S_{\mathrm{cond}}(k) = \left\langle \left[\frac{\log P_a^{\mathrm{gen}}(k;\theta) - \mu_a^{\mathrm{true}}(k;\theta)}{\sigma_a^{\mathrm{true}}(k;\theta)}\right]^2 \right\rangle_\theta, \tag{17}$$

where μ_a^{true} and σ_a^{true} are computed from the 15 CAMELS projections at the same conditioning vector. S_{cond} is used as a diagnostic of within-condition ensemble spread rather than as a formal calibration test. Since it is computed relative to the CAMELS within-condition mean, it includes both generated scatter and any residual mean offset. Values near unity indicate that the normalized RMS deviation of the generated samples is comparable to the CAMELS projection-level scatter.

The left panel of Figure 6 shows that GENESIS tracks a substantial fraction of the parameter-dependent variation in the log-power spectra. The overall R^2 values are 0.842, 0.882, and 0.833 for M_{cdm}, M_{gas}, and T, respectively. To anchor these numbers, a parameter-independent mean predictor that always returns the LH-averaged log-power spectrum regardless of $\boldsymbol{\theta}$ would have $R^2 = 0$, while perfect tracking would give $R^2 = 1$. Thus, by this metric, the conditioning in GENESIS recovers approximately 83–88 per cent of the variance in log-power across the LH parameter space.

Interestingly, the highest R^2 is obtained for M_{gas} rather than for the gravitationally simpler M_{cdm} channel. This likely reflects the stronger parameter response of the gas-density field across the six-dimensional CAMELS design space. Supernova and AGN feedback variations introduce substantial variation in $\log P_{\mathrm{gas}}(k;\boldsymbol{\theta})$, increasing the denominator $\mathrm{Var}_\theta[\log P_a^{\mathrm{true}}(k;\theta)]$ in equation (16). By contrast, the dark-matter field is sensitive primarily to Ω_{m} and σ_8 and only weakly to the feedback parameters, so its conditional-variance denominator is smaller. Comparable model errors can therefore lead to a slightly lower R^2 for M_{cdm}.

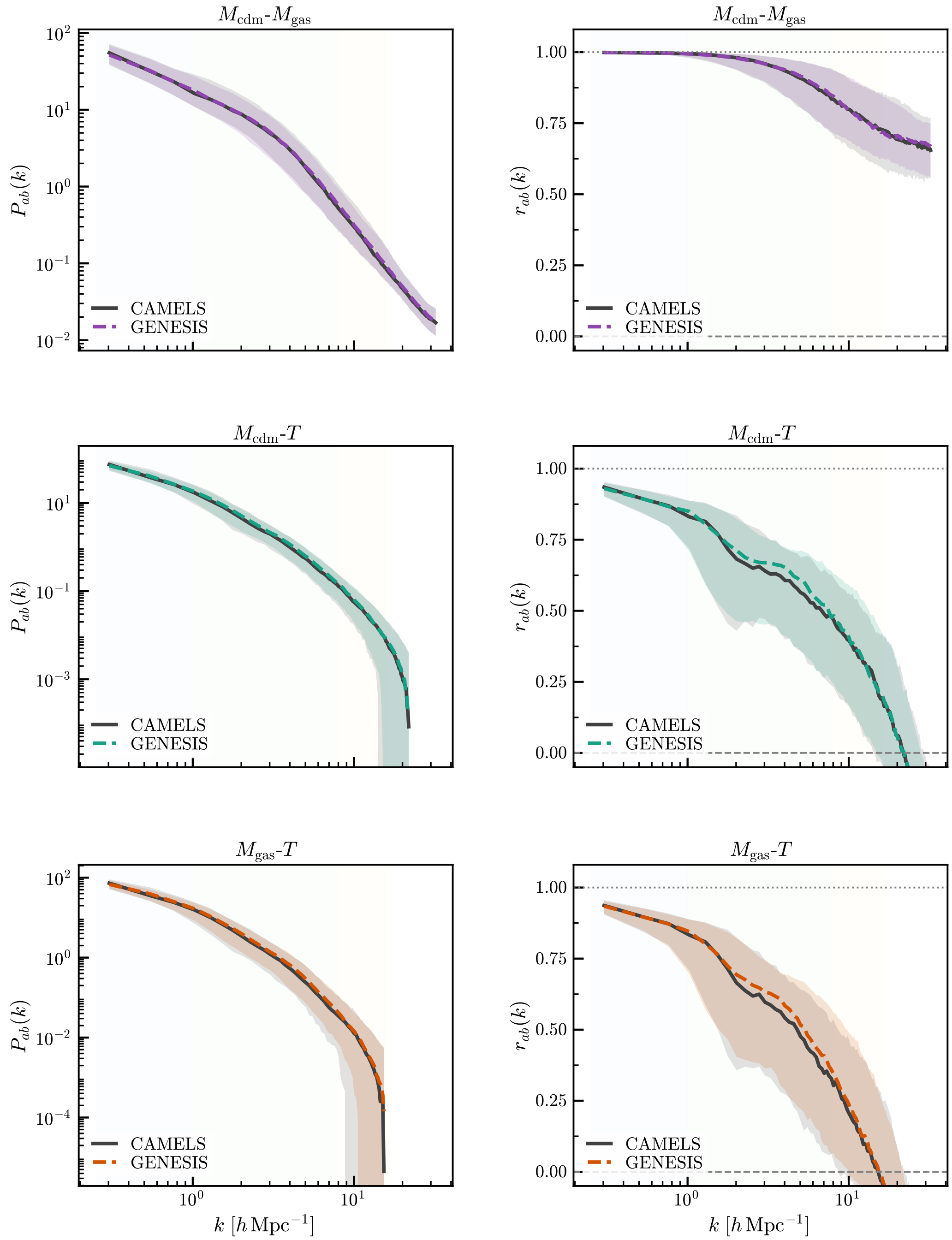

Figure 5: Cross-power spectra and inter-field coherence on the LH test set. Left panels show $P_{ab}(k)$ and right panels show the coherence $r_{ab}(k)$ for the three field pairs $M_{\rm cdm}$–$M_{\rm gas}$, $M_{\rm cdm}$–T, and $M_{\rm gas}$–T. Black curves show CAMELS statistics and dashed coloured curves show GENESIS; shaded regions indicate the 16–84 percentile ranges over the held-out LH set.

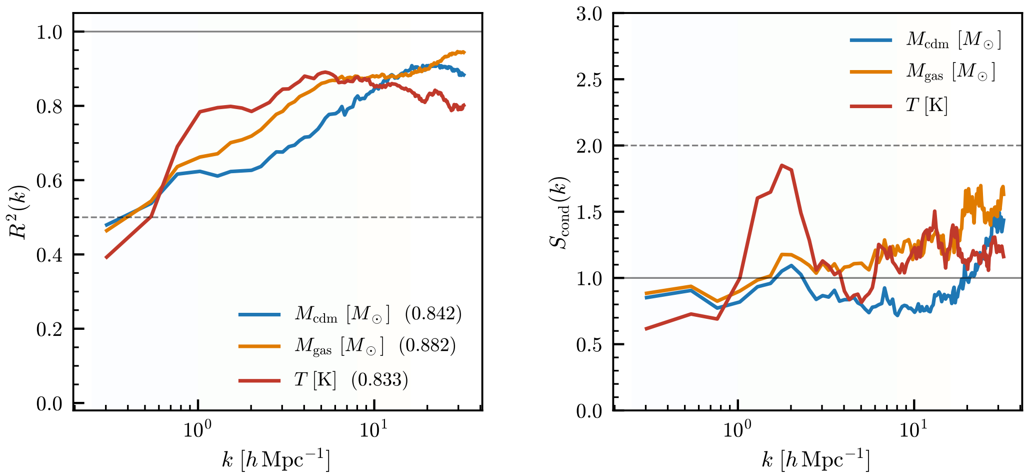

Figure 6: Conditional response and ensemble spread on the LH test set. Left: $R^2(k)$ of the generated log-power response across the held-out conditioning vectors. Right: $S_{\rm cond}(k)$, used here as a within-condition spread diagnostic.

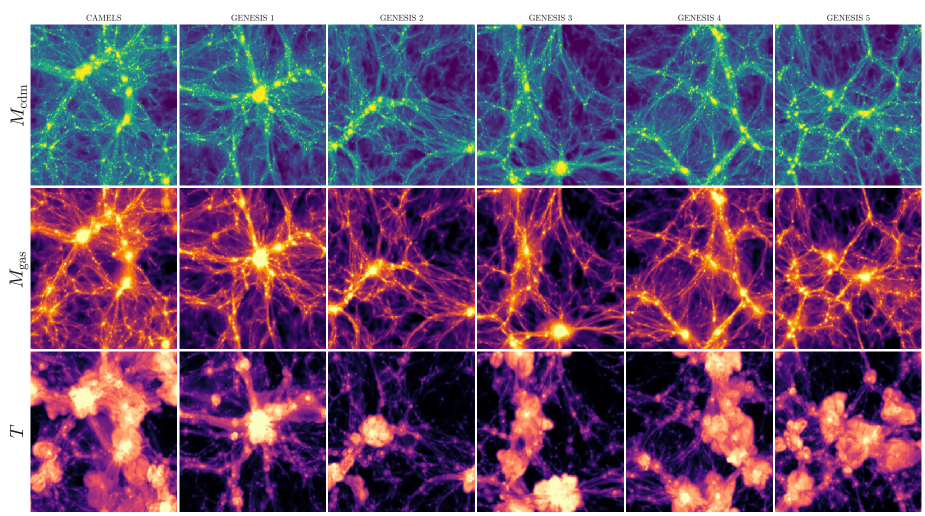

Figure 7: Comparison between CAMELS CV maps and GENESIS-generated maps, conditioned on $(\Omega_{\rm m}, \sigma_8, A_{\rm SN1}, A_{\rm SN2}, A_{\rm AGN1}, A_{\rm AGN2}) = (0.30, 0.80, 1, 1, 1, 1)$. CAMELS realizations are compared with independent GENESIS samples generated at the same conditioning vector. Rows show $M_{\rm cdm}$, $M_{\rm gas}$, and T.

The right panel of Figure 6 shows that $S_{\rm cond}$ remains broadly of order unity for much of the plotted range, especially for $M_{\rm cdm}$. For the baryonic fields, $S_{\rm cond}$ increases on feedback-sensitive scales, indicating larger normalized RMS deviations from the CAMELS within-condition mean. Values of $S_{\rm cond} \sim 2$ correspond to RMS deviations of order $\sqrt{2} \simeq 1.4$ times the CAMELS projection-level scatter, although this quantity includes both generated scatter and any residual mean offset. The largest departures are visible for $M_{\rm gas}$ and T, where feedback and thermal processes introduce additional stochasticity. This behaviour is interpreted as mild over-dispersion or residual bias in the most feedback-sensitive regimes, rather than as a formal coverage statement.

Overall, the LH evaluation shows that GENESIS reproduces the main per-field and cross-field statistics of CAMELS multifield maps across the six-dimensional parameter space. The strongest agreement appears in the auto-power spectra, one-point distributions, per-simulation field means, and conditional

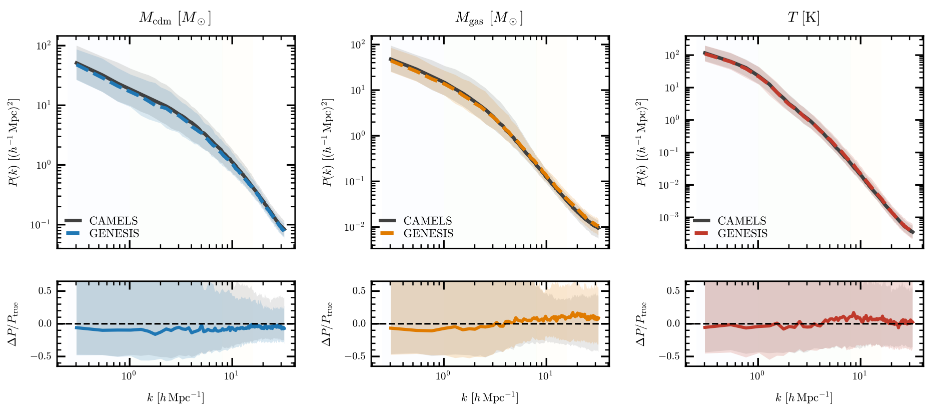

Figure 8: Auto-power spectra on the CV set. Top panels show CAMELS and GENESIS auto-power spectra for $M_{\rm cdm}$, $M_{\rm gas}$, and T; shaded regions indicate the 16–84 percentile ranges. Bottom panels show fractional residuals relative to the CAMELS median.

log-power response. The largest remaining discrepancies occur in feedback-sensitive baryonic statistics, especially temperature-related correlations and small-scale ensemble spread. Additional LH map tiles across a wider range of conditioning vectors are shown in Appendix C.

5 CV Evaluation

This section evaluates GENESIS on the CAMELS/CMD CV set, where all simulations share the same parameter vector $\boldsymbol{\theta}$ but differ in initial conditions. Section 5.1 presents representative generated map triplets at the fiducial parameter point, Section 5.2 compares the auto-power spectra, Section 5.3 examines CV-normalized spectral displacement and map-level spread, Section 5.4 compares one-point field distributions, and Section 5.5 evaluates cross-power spectra and inter-field coherence. For the CV comparisons, 200 independent GENESIS samples are generated at the fiducial conditioning vector; this takes roughly 20 minutes on a single NVIDIA A100 GPU in the current implementation.

5.1 Field morphology at fixed parameters

A visual comparison of GENESIS maps from the CV set is first presented, as in the LH evaluation. Since the 27 CV simulations themselves differ in initial conditions, the relevant question is whether samples generated at the fiducial conditioning vector lie within the morphological range spanned by independent CAMELS realizations.

Figure 7 shows that GENESIS produces visually plausible samples at the fiducial parameter point. The dark-matter and gas-density channels show filamentary structure and gas distributions that fall within the visual range of independent CAMELS realizations. The temperature channel again shows softer small-scale features in the generated samples, consistent with the small-scale over-dispersion quantified in Section 5.3. The remaining subsections test this agreement more quantitatively using summary statistics across the full CV set.

5.2 Auto-power spectra

The auto-power spectra of the generated maps are first compared with those of the CAMELS CV maps. Since the CV set is evaluated at fixed parameters, this comparison tests whether GENESIS produces samples with similar spectral statistics to independent CAMELS realizations at the fiducial parameter point.

Figure 8 shows that the generated spectra follow the CAMELS medians for all three fields. The residuals are generally small compared with the map-level scatter, although mild scale-dependent differences remain. For $M_{\rm cdm}$, the residual is slightly below zero over most of the plotted range, indicating a small low-power bias in the generated dark-matter spectra. The baryonic fields show

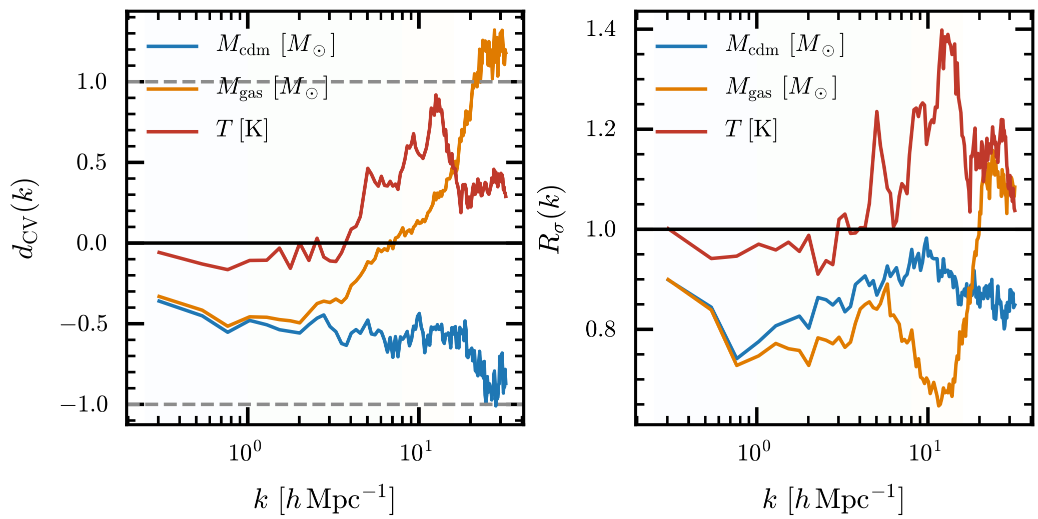

Figure 9: CV-normalized spectral displacement and map-level spread. Left: $d_{\rm CV}(k)$ for the generated mean auto-power spectrum. Right: $R_\sigma(k)$, the ratio of generated to CAMELS map-level log-power variance.

additional scale-dependent departures, as expected from their stronger sensitivity to feedback and thermal processes. Overall, the generated samples reproduce the main two-point clustering amplitudes at the fiducial CV parameter point, with these residual trends indicating mild field-dependent biases rather than large failures.

5.3 CV-normalized displacement and map-level spread

The spectral comparison is further summarized using a CV-normalized displacement,

$$d_{{\rm CV},a}(k) = \frac{\langle P_a^{\rm gen}(k)\rangle_{\rm gen} - \langle \overline{P}_{a,j}^{\rm CV}(k)\rangle_j}{{\rm std}_j[\overline{P}_{a,j}^{\rm CV}(k)]}, \tag{18}$$

where $\overline{P}_{a,j}^{\rm CV}(k)$ denotes the mean auto-power spectrum of the 15 projected maps from CV simulation j. This quantity measures the displacement of the generated mean spectrum in units of the simulation-to-simulation scatter among the CV runs.

A map-level spread ratio is also considered,

$$R_{\sigma,a}(k) = \frac{{\rm Var}_{\rm gen}[\log P_a(k)]}{{\rm Var}_{\rm CV,map}[\log P_a(k)]}. \tag{19}$$

R_σ is used only as a diagnostic of relative spread. The CAMELS CV maps include both realization-to-realization variation and projection-level scatter, while the generated maps are independent samples from the learned conditional model.

Figure 9 shows that $d_{\rm CV}$ provides a directional measure of bias at the fiducial parameter point. The dark-matter channel exhibits a coherent $d_{\rm CV} \simeq -0.5$ across most scales, indicating that the GENESIS mean spectrum lies roughly half of the CV simulation-to-simulation scatter below the CAMELS mean. This is a systematic negative bias, not random scatter: the displacement is coherent across the full k-range, which indicates a consistent amplitude offset in the generated dark-matter power. The gas-density channel reaches $d_{\rm CV} \simeq -1$ on some scales — approaching the full magnitude of the intrinsic CV scatter — and, critically, the bias is again coherent in sign across scales rather than fluctuating around zero. Although these values nominally fall within one standard deviation of the CV ensemble, the consistent directional bias across scales is a strong indicator that the model carries a systematic offset, not statistical noise. The temperature channel remains closer to unbiased, with $d_{\rm CV}$ varying around zero and becoming mildly positive toward high k.

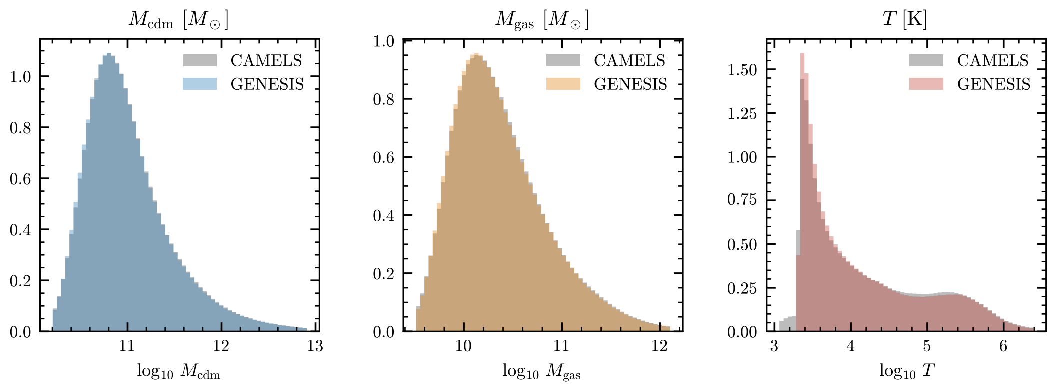

Figure 10: One-point pixel distributions on the CV set in $\log_{10}$ field space. Grey histograms show CAMELS and coloured histograms show GENESIS.

The map-level spread ratio R_σ gives a complementary view of the generated ensemble. For $M_{\rm cdm}$, $R_\sigma \simeq 0.8$, indicating that the generated samples are somewhat under-dispersed relative to the CAMELS map-to-map variation. For $M_{\rm gas}$, R_σ remains close to unity, showing well-matched map-level spread. For T, R_σ increases to approximately 1.4 on small scales, indicating systematic over-dispersion of temperature fluctuations where feedback and thermal processes dominate. This small-scale over-dispersion is consistent with the elevated $S_{\rm cond}$ values seen for the LH set in Section 4.5. Together, the two CV diagnostics reveal a model that carries systematic biases: a coherent negative amplitude offset in dark matter, and a systematic over-dispersion in temperature on small scales — both of which would propagate into downstream parameter inference if left uncorrected.

5.4 One-point field distributions

One-point pixel distributions in $\log_{10}$ field space are next compared. This complements the power-spectrum comparison by testing the absolute range of field values rather than only relative fluctuations in overdensity.

Figure 10 shows that the CAMELS and GENESIS pixel distributions are similar for all three fields. The agreement is particularly close for the two density fields. The temperature distribution is broader and more structured, but the generated samples still follow its main shape. This suggests that the model captures the typical field amplitudes at the fiducial parameter point, while leaving some differences in the more feedback-sensitive temperature field.

5.5 Cross-power spectra and inter-field coherence

Finally, the joint structure of the three generated fields is examined. The cross-power spectra $P_{ab}(k)$ and coherence $r_{ab}(k)$ for the field pairs $M_{\rm cdm}$–$M_{\rm gas}$, $M_{\rm cdm}$–T, and $M_{\rm gas}$–T are compared.

Figure 11 shows that the generated samples are broadly consistent with the CAMELS cross-field statistics. The $M_{\rm cdm}$–$M_{\rm gas}$ pair shows the closest agreement, while the temperature-related pairs show larger differences, particularly on smaller scales. This is consistent with the greater sensitivity of temperature to shock heating and feedback processes. Overall, the comparison indicates that GENESIS preserves the main multifield correlations at the fiducial parameter point, with the largest residuals appearing in the baryonic and temperature-related statistics.

Taken together, the CV diagnostics show that GENESIS produces samples with broadly similar auto-power spectra, one-point distributions, and cross-field correlations to CAMELS at the fiducial parameter vector. The comparison also identifies remaining differences in map-level spread and in temperature-related baryonic statistics. The 200-sample CV ensemble is generated in roughly 20 minutes on a single NVIDIA A100 GPU, indicating that forward sampling is practical for moderate ensemble sizes even though likelihood evaluation remains substantially more expensive.

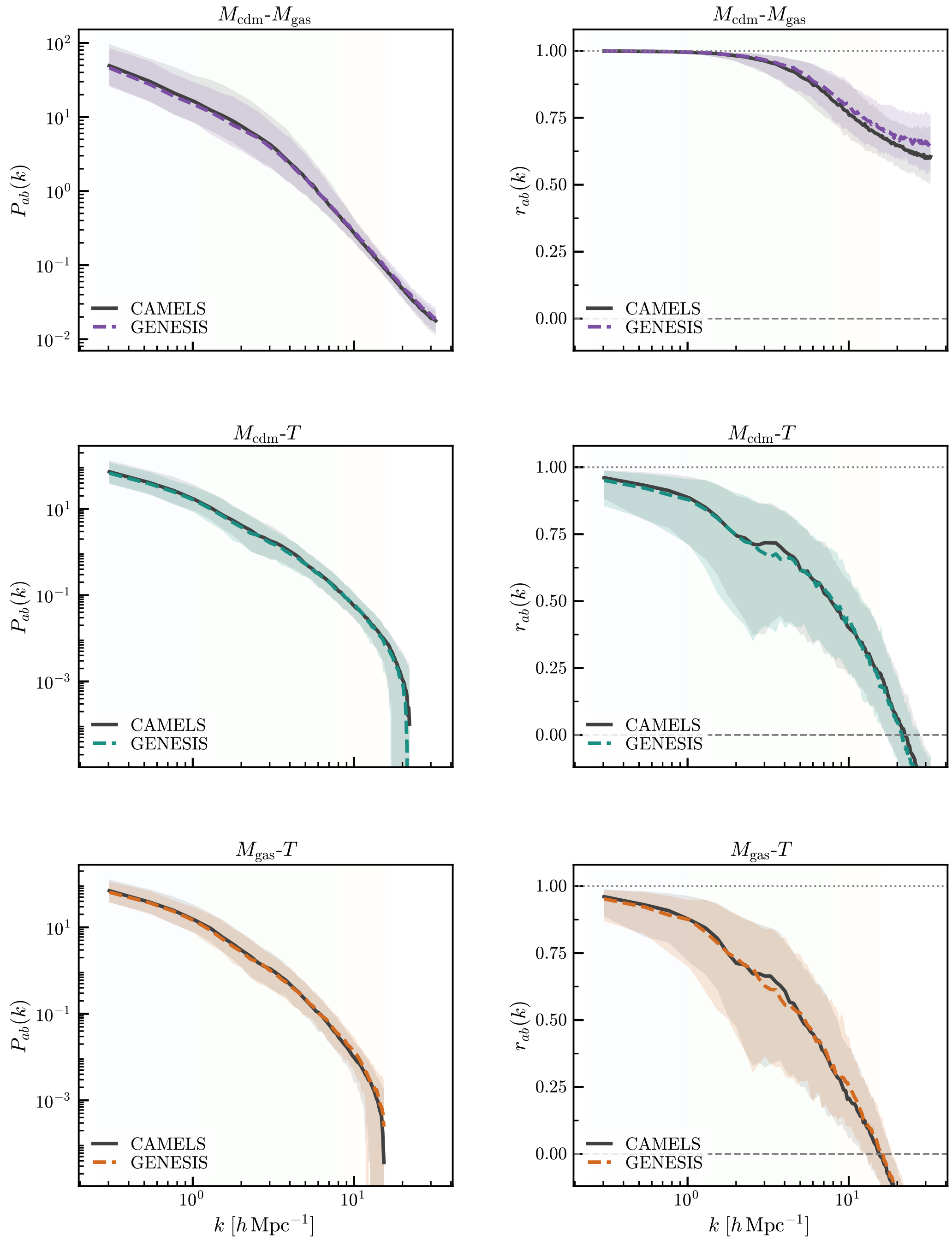

Figure 11: Cross-power spectra and inter-field coherence on the CV set. Left panels show $P_{ab}(k)$ and right panels show the coherence $r_{ab}(k)$ for the three field pairs. Black curves show CAMELS and dashed coloured curves show GENESIS; shaded regions indicate 16–84 percentile ranges.

6 Likelihood-based parameter inference

This section demonstrates likelihood-based parameter inference with GENESIS by fixing one observed CAMELS multifield map, denoted $\mathbf{x}_1$, and scanning the cosmological parameters Ω_m and σ_8. Section 6.1 describes likelihood evaluation through the change-of-variables formula for continuous normalizing flows, Section 6.2 describes the stochastic trace estimator and summarizes the scan setup, and Section 6.3 presents a two-dimensional profile likelihood scan in the Ω_m–σ_8 plane.

6.1 Likelihood evaluation by change of variables

GENESIS defines an ODE transport from the base distribution to the conditional data distribution. As a result, likelihoods can be evaluated using the instantaneous change-of-variables formula for continuous normalizing flows [16]. In the likelihood scan, the observed map $\mathbf{x}_1$ is fixed and only the candidate conditioning vector $\boldsymbol{\theta}$ is varied. The learned probability-flow ODE is

$$\frac{\mathrm{d}\mathbf{x}_t}{\mathrm{d}t} = u_t^\phi(\mathbf{x}_t \mid \boldsymbol{\theta}), \qquad t \in [0,1], \tag{20}$$

with $\mathbf{x}_0 \sim p_0 = \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\mathbf{x}_1$ in the data distribution. Along the forward flow, the instantaneous change in log density is

$$\frac{\mathrm{d}}{\mathrm{d}t} \log p_t(\mathbf{x}_t \mid \boldsymbol{\theta}) = -\nabla_\mathbf{x} \cdot u_t^\phi(\mathbf{x}_t \mid \boldsymbol{\theta}). \tag{21}$$

For likelihood evaluation, the ODE is solved in reverse from the fixed data point $\mathbf{x}_1$ to the corresponding latent point $\mathbf{z}_0$ in the noise space. Equivalently, writing the associated forward trajectory from $\mathbf{z}_0$ to $\mathbf{x}_1$, the conditional log likelihood is

$$\log p_\phi(\mathbf{x}_1 \mid \boldsymbol{\theta}) = \log p_0(\mathbf{z}_0) - \int_0^1 \nabla_{\mathbf{x}_t} \cdot u_t^\phi(\mathbf{x}_t \mid \boldsymbol{\theta}) \, \mathrm{d}t. \tag{22}$$

In practice, the same quantity is accumulated during reverse ODE integration with the corresponding sign convention. Relative likelihoods are reported,

$$\Delta \log p(\boldsymbol{\theta}) = \log p_\phi(\mathbf{x}_1 \mid \boldsymbol{\theta}) - \max_{\boldsymbol{\theta}'} \log p_\phi(\mathbf{x}_1 \mid \boldsymbol{\theta}'), \tag{23}$$

where the maximum is taken over the evaluated scan grid. Thus the best grid point has $\Delta \log p = 0$, and all other grid points have negative values.

6.2 Stochastic trace estimation and scan setup

The divergence term $\nabla_{\mathbf{x}_t} \cdot u_t^\phi(\mathbf{x}_t \mid \boldsymbol{\theta})$ in equation (22) is the trace of the velocity-field Jacobian,

$$\nabla_{\mathbf{x}_t} \cdot u_t^\phi(\mathbf{x}_t \mid \boldsymbol{\theta}) = \mathrm{Tr}\left[\frac{\partial u_t^\phi}{\partial \mathbf{x}_t}\right]. \tag{24}$$

Direct evaluation of this trace would require the full Jacobian at each ODE step. For the three 256×256 fields considered here, this Jacobian acts on all pixels and channels, making explicit trace evaluation computationally prohibitive. Following standard CNF likelihood practice [38, 15], the divergence is therefore estimated stochastically using the Hutchinson trace estimator [39],

$$\mathrm{Tr}\left[\frac{\partial u_t^\phi}{\partial \mathbf{x}_t}\right] = \mathbb{E}_{\boldsymbol{\epsilon}}\left[\boldsymbol{\epsilon}^\mathsf{T} \frac{\partial u_t^\phi}{\partial \mathbf{x}_t} \boldsymbol{\epsilon}\right], \qquad \mathbb{E}[\boldsymbol{\epsilon}\boldsymbol{\epsilon}^\mathsf{T}] = \mathbf{I}. \tag{25}$$

Rademacher probe vectors are used in the calculations below, with each component independently taking values ± 1 with equal probability. The likelihood values are averaged over independent Hutchinson estimates to reduce trace-estimator noise.

For the proof-of-concept scan, only the two cosmological parameters Ω_m and σ_8 are varied. The four astrophysical feedback parameters are fixed to the CAMELS values of the selected map, and candidate cosmologies are evaluated on a local grid around the true parameter point. The scan is restricted to a broad ellipse around this region to avoid unnecessary evaluations far from the local likelihood peak. The numerical scan settings are listed in Appendix B.

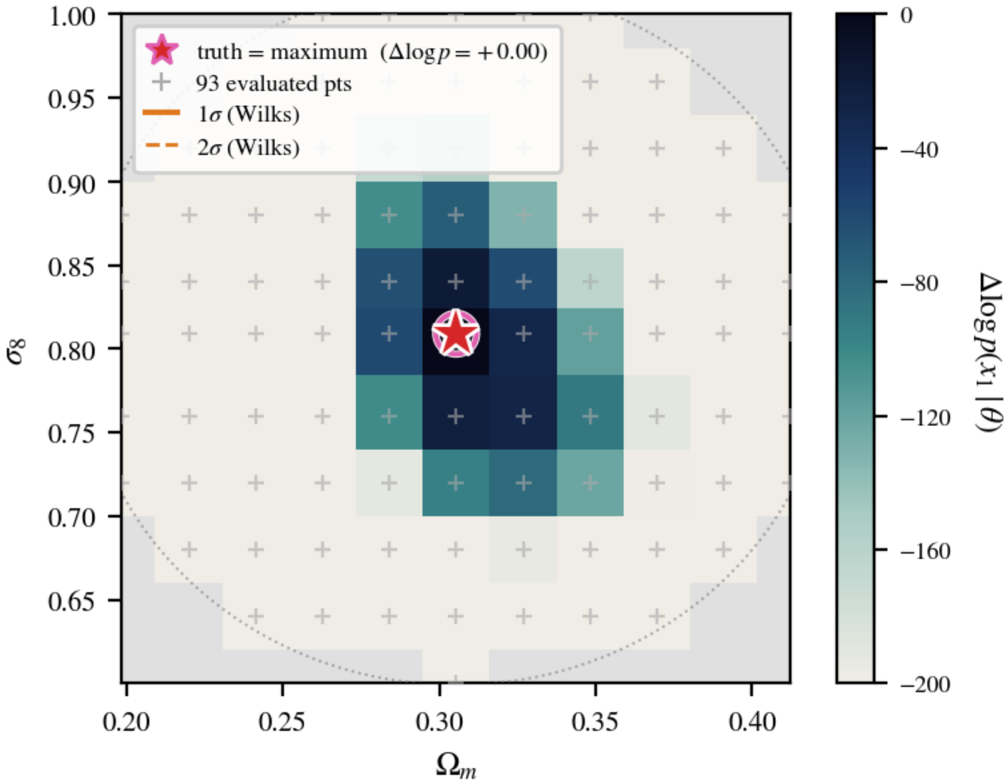

Figure 12: Two-dimensional likelihood scan in the $\Omega_{\rm m}$–σ_8 plane for a fixed CAMELS map $\mathbf{x}_1$. The colour scale shows $\Delta \log p(\mathbf{x}_1 \mid \boldsymbol{\theta})$ relative to the maximum evaluated grid value. The true CAMELS parameter and the maximum-likelihood grid point coincide. Grey regions indicate grid points outside the evaluated scan ellipse, and markers show evaluated grid points. Contours show the Wilks-theorem 1 and 2σ levels for two parameters.

6.3 Two-dimensional profile likelihood scan

Figure 12 shows that the seed-averaged relative likelihood peaks at the true CAMELS parameter point, which is included explicitly in the scan grid. The colour scale shows the relative log likelihood, not an absolute likelihood; values close to zero therefore correspond to conditioning vectors that best explain the fixed map $\mathbf{x}_1$ within the evaluated grid. The surface is broad and the grey region marks grid points outside the evaluated region, so this result should be interpreted as a proof-of-concept likelihood scan rather than a precise cosmological constraint.

The profile projections in Figure 13 show the same qualitative behaviour. For each value of one parameter, the profile keeps the largest relative likelihood over the other parameter,

$$\begin{aligned} \Delta \log p_{\rm prof}(\Omega_{\rm m}) &= \max_{\sigma_8} \Delta \log p(\Omega_{\rm m}, \sigma_8), \\ \Delta \log p_{\rm prof}(\sigma_8) &= \max_{\Omega_{\rm m}} \Delta \log p(\Omega_{\rm m}, \sigma_8). \end{aligned} \tag{26}$$

Both profiles peak at the true CAMELS value and remain broad, with visible uncertainty from the stochastic trace estimator. This result shows that the learned flow likelihood contains useful parameter information for a low-dimensional scan on CAMELS data. A full inference analysis would require sampling or scanning over the remaining astrophysical parameters, more likelihood evaluations, and calibration over multiple maps.

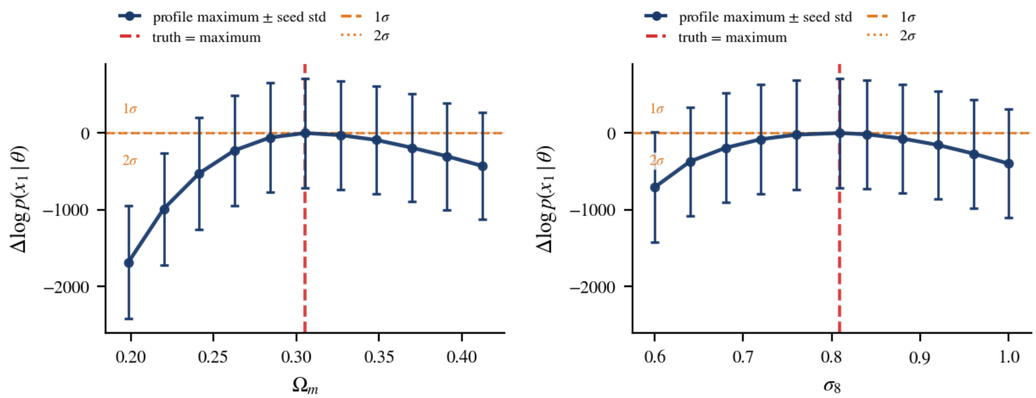

Figure 13: One-dimensional profile projections of the likelihood scan. Left: profile in Ω_m, obtained by maximizing over σ_8 at each Ω_m. Right: profile in σ_8, obtained by maximizing over Ω_m at each σ_8. Error bars show the scatter across Hutchinson seeds. The vertical dashed line marks the CAMELS parameter value. Horizontal lines show the 1D Wilks-theorem 1 and 2σ levels.

7 Discussion and Further Work

The evaluations above show that GENESIS can generate multifield CAMELS maps whose summary statistics are broadly similar to those of the target simulations. Across the LH test set, the model matches the marginal auto-power spectra, one-point field distributions, per-simulation mean amplitudes, and inter-field correlations at the level of the diagnostics considered in this work. The CV comparison gives a complementary fixed-parameter check, showing that generated samples at the fiducial parameter point have similar auto-power spectra, pixel distributions, and cross-field statistics to independent CAMELS realizations.

At the same time, the residuals are not merely noise: they reflect systematic biases that must be taken seriously. The CV diagnostics show that the dark-matter channel has a coherent $d_\mathrm{CV} \simeq -0.5$ across scales, and the gas-density channel reaches $d_\mathrm{CV} \simeq -1$ in some k-bins. Although both values fall nominally within one standard deviation of the CV ensemble, the bias is consistently negative across the full scale range for both channels — it does not fluctuate around zero as random error would. This pattern is a clear signature of a systematic model bias, not sampling variance. A model that persistently underestimates the clustering amplitude by a coherent fraction of the cosmic-variance scatter would, if used for parameter inference without correction, produce systematically biased parameter estimates.

The cross-field diagnostics are particularly important for interpreting the model as a multifield emulator. A model could reproduce the auto-power spectra of M_cdm, M_gas, and T separately while still generating mutually inconsistent fields. The cross-power and coherence results indicate that GENESIS preserves the main spatial coupling among the three fields. The agreement is strongest for the M_cdm–M_gas pair, while temperature-related pairs show larger residuals, especially toward smaller scales. This behaviour is consistent with the temperature field being more sensitive to local heating and feedback processes than to density alone.

The likelihood scan demonstrates a separate consequence of the CNF formulation: the same trained model can be used to assign likelihoods to input maps under candidate conditioning vectors. In the two-dimensional Ω_m–σ_8 scan, one CAMELS map $\mathbf{x}_1$ is held fixed while the conditioning vector is varied over a local grid around the true parameter point. The four astrophysical feedback parameters are fixed to their CAMELS values, and the relative likelihood is evaluated only in the neighbourhood targeted by the scan. The maximum-likelihood grid point coincides with the truth, and the Wilks-theorem 1σ contour encloses several adjacent grid points. This result is interpreted as a proof of concept that the likelihood surface is centred on the truth at the resolution of the scan, rather than that the truth is uniquely resolved within stochastic noise.

There are several limitations to this likelihood-based inference demonstration. First, the scan varies only two cosmological parameters and keeps the four astrophysical feedback parameters fixed. It is therefore a low-dimensional profile likelihood slice, not a full posterior analysis over the six-dimensional CAMELS parameter space. In addition, the finite grid and the local scan region limit the interpretation of the likelihood contours. The scan is therefore an encouraging consistency check on the local likelihood surface, but not a standalone validation of calibrated inference.

Second, likelihood evaluation is computationally expensive. In the scan presented here, each conditioning vector requires an ODE solve with stochastic divergence estimation. This makes dense scans or marginalization over the full CAMELS parameter space substantially more costly than forward generation. The Hutchinson trace estimator is unbiased and gives a consistent relative trend across the scan, but it still introduces stochastic noise into individual likelihood evaluations. Reducing this noise requires additional trace estimates, which directly increases the runtime of each grid point. Variance-reduction strategies, improved ODE solvers, or learned approximations to the likelihood surface may therefore be required for practical inference applications.

A broader statistical limitation is that the current evaluation does not yet provide a fully calibrated inference procedure. The LH and CV diagnostics test whether generated fields reproduce selected summary statistics, and the likelihood scan checks whether the CNF likelihood peaks near the correct region for one CAMELS map. These tests are useful, but they do not by themselves establish calibrated parameter coverage, unbiased posterior means, or reliable credible intervals. Such claims would require repeated inference experiments over many maps, posterior calibration tests, and marginalization over all relevant cosmological and astrophysical parameters.

The present work is also limited to two-dimensional projected maps from a single simulation suite. Projection removes line-of-sight information, and the $25\,h^{-1}\mathrm{Mpc}$ CAMELS volume limits the available large-scale modes. Extending the approach to larger volumes or three-dimensional fields would allow a more direct connection to survey-scale clustering statistics, but would also require more memory efficient architectures and more expensive validation. In addition, GENESIS is trained only on the IllustrisTNG branch of CAMELS. Generalization to other hydrodynamic models, such as SIMBA or EAGLE, would require retraining, transfer learning, or explicit multi-suite training to separate cosmological dependence from baryonic model dependence.

These limitations suggest several natural next steps. Using the CAMELS LH test set and the 1P set, future validation will test whether GENESIS correctly tracks individual one-parameter response curves for each of the six conditioning parameters in isolation. The EX set will then be used to probe robustness in extreme conditions near or beyond the edges of the training range. Beyond validation, a natural direction is to broaden the scope of the model: extending training to multiple redshift snapshots, to three-dimensional field grids, and to other simulation suites such as SIMBA, which would allow the model to generalise beyond the IllustrisTNG baryonic prescription and across cosmic time. For likelihood-based inference, future work will evaluate the CNF likelihood across many CAMELS maps, scan or sample the full six-dimensional parameter space, and quantify estimator variance. More importantly, rather than restricting inference to grid scans, Hamiltonian Monte Carlo and energy-based sampling will be explored to draw full posterior samples — thereby testing the true capacity of GENESIS as a field-level parameter inference engine.

8 Conclusions

GENESIS has been presented as a conditional generative emulator for multifield cosmological maps based on optimal-transport flow matching. The model jointly generates dark matter density, gas density, and gas temperature fields, conditioned on the full six-dimensional CAMELS parameter vector. The model was evaluated on both the parameter-varying LH test set and the fixed-parameter CV set using a suite of one-point, two-point, cross-field, and conditional diagnostics.

Across these tests, GENESIS reproduces the main CAMELS statistics with good overall fidelity. The generated maps match the marginal auto-power spectra, one-point distributions, field means, and cross-field structure reasonably well, and the conditional log-power response tracks much of the variation across the LH parameter space. The remaining discrepancies, however, are more than incidental scatter. The temperature field exhibits systematic over-dispersion by a factor of $R_\sigma \simeq 1.4$ on small scales, indicating that the model does not correctly reproduce the conditional variance of the most feedback-sensitive channel. The CV diagnostics reveal a coherent negative bias of $d_{\mathrm{CV}} \simeq -0.5$ in the dark-matter channel — a systematic underprediction of roughly half the cosmic-variance scatter that would bias downstream parameter inference if left uncorrected. These discrepancies indicate that further calibration is required before precision inference applications. In the current implementation, generating 200 independent multifield samples at a fixed conditioning vector takes roughly 20 minutes on a single NVIDIA A100 GPU, showing that moderate conditional ensembles are feasible.

Finally, a proof-of-concept likelihood evaluation has been demonstrated. In a two-dimensional Ω_{m}–σ_8 scan with the other parameters fixed, the likelihood maximum coincides with the true CAMELS parameter value for the fixed observed map. This is not yet a calibrated cosmological inference pipeline, but it suggests that the learned flow likelihood contains useful parameter information. Overall, these results indicate that flow matching can be a promising tool for both conditional cosmological field generation and field-level likelihood-based inference.

Acknowledgements

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Data Availability

The CAMELS simulation data used in this work are publicly available at `https://camels.readthedocs.io`. The trained GENESIS model weights, training code, evaluation pipeline, and scripts to reproduce all figures in this paper will be made available at `https://github.com/pmj0324/GENESIS` upon publication.

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A Training details

GENESIS is trained in three stages. The initial training stage uses AdamW [37] with learning rate $\eta = 2 \times 10^{-4}$, weight decay 10^{-2}, and $(\beta_1, \beta_2) = (0.9, 0.999)$. The learning rate is warmed up for 5 epochs and then controlled by a ReduceLROnPlateau scheduler. This stage is run for 89 epochs. Two fine-tuning stages are then performed: the first uses a smaller learning rate, $\eta = 2 \times 10^{-5}$, no warm-up, and a ReduceLROnPlateau scheduler with factor 0.65 and patience 4. This stage is run for 42 epochs. The final fine-tuning stage starts from the first fine-tuned model and disables classifier-free conditioning dropout by setting $p_{\mathrm{cfg}} = 0$. This stage is run for 25 epochs and provides the checkpoint used for all results reported in this work. All stages use batches of $B = 16$ multifield map triplets and D4 data augmentation. Training is performed on a single NVIDIA A100 GPU.

B Likelihood scan details

The likelihood scan in Section 6 uses one CAMELS map whose true conditioning vector is

$$\boldsymbol{\theta}_{\mathrm{true}} = (0.3054,\ 0.8090,\ 3.7373,\ 3.8106,\ 1.3822,\ 0.6639). \tag{27}$$

The first two entries are close to the Planck 2018 cosmology. The candidate grid is

$$\begin{aligned} \Omega_{\mathrm{m}} \in \{&0.1985, 0.2199, 0.2413, 0.2626, 0.2840, 0.3054,\\ &0.3268, 0.3482, 0.3695, 0.3909, 0.4123\}, \end{aligned} \tag{28}$$

$$\begin{aligned} \sigma_8 \in \{&0.6000, 0.6400, 0.6800, 0.7200, 0.7600, 0.8090,\\ &0.8400, 0.8800, 0.9200, 0.9600, 1.0000\}. \end{aligned} \tag{29}$$

The true cosmological point is included explicitly in the grid. To avoid evaluating points far from the region of interest, the calculation is restricted to

$$\left(\frac{\Omega_{\mathrm{m}} - \Omega_{\mathrm{m,true}}}{0.04}\right)^2 + \left(\frac{\sigma_8 - \sigma_{8,\mathrm{true}}}{0.07}\right)^2 \le 3^2. \tag{30}$$

This cut removes 28 of the 121 candidate grid points, leaving 93 evaluated likelihoods. A Dormand–Prince solver is used, with 100 function evaluations per likelihood.

Rademacher probe vectors are used for the Hutchinson trace estimate. For each evaluated grid point, three independent Hutchinson estimates are computed and averaged as

$$\overline{\log p}_{ij} = \frac{1}{3} \sum_{s=0}^{2} \log p_{\phi,s}(\mathbf{x}_1 \mid \boldsymbol{\theta}_{ij}). \tag{31}$$

The scatter across the three estimates is used as the uncertainty shown in Figure 13. In the current implementation, one likelihood evaluation takes roughly two minutes on an NVIDIA A100 GPU.

C Additional LH map tiles

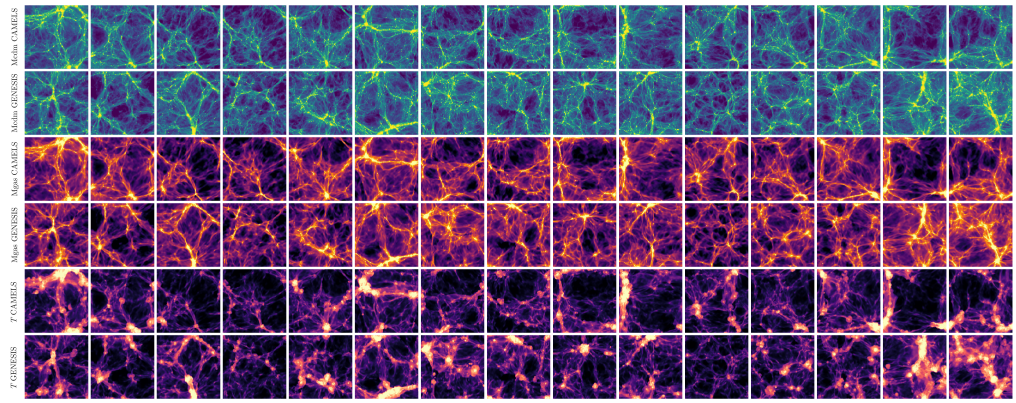

Figure 14. LH 0719: $(\Omega_\mathrm{m}, \sigma_8, A_\mathrm{SN1}, A_\mathrm{SN2}, A_\mathrm{AGN1}, A_\mathrm{AGN2}) = (0.104, 0.747, 3.308, 0.446, 1.461, 0.549)$. Each tile arranges 15 generated samples with the corresponding CAMELS reference rows.

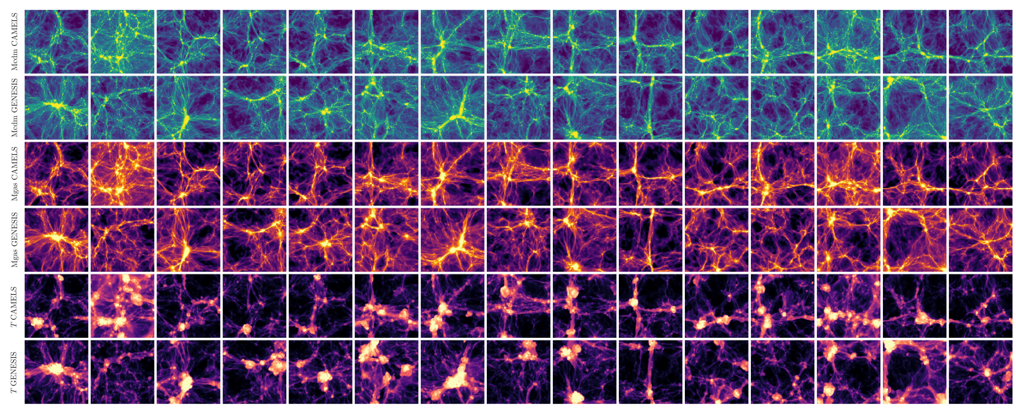

Figure 15. LH 0218: $(\Omega_\mathrm{m}, \sigma_8, A_\mathrm{SN1}, A_\mathrm{SN2}, A_\mathrm{AGN1}, A_\mathrm{AGN2}) = (0.150, 0.743, 0.924, 0.342, 1.030, 0.523)$. Each tile arranges 15 generated samples with the corresponding CAMELS reference rows.

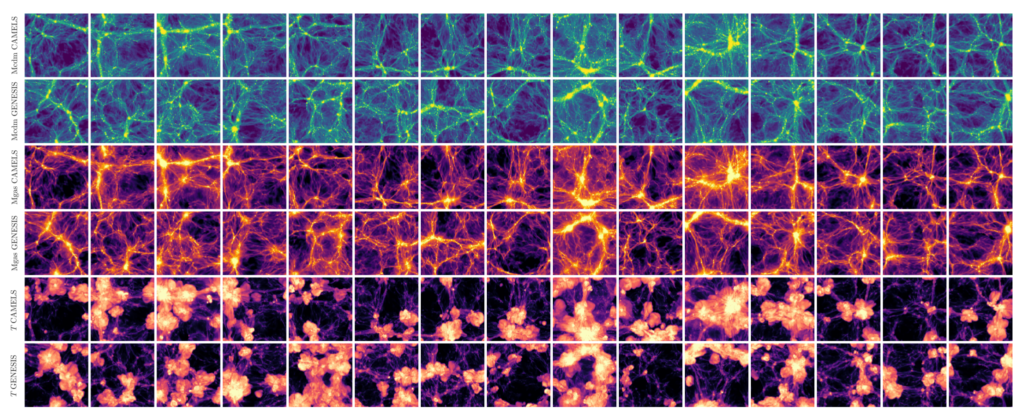

Figure 16. LH 0015: $(\Omega_\mathrm{m}, \sigma_8, A_\mathrm{SN1}, A_\mathrm{SN2}, A_\mathrm{AGN1}, A_\mathrm{AGN2}) = (0.218, 0.788, 0.825, 0.274, 0.552, 1.106)$. Each tile arranges 15 generated samples with the corresponding CAMELS reference rows.

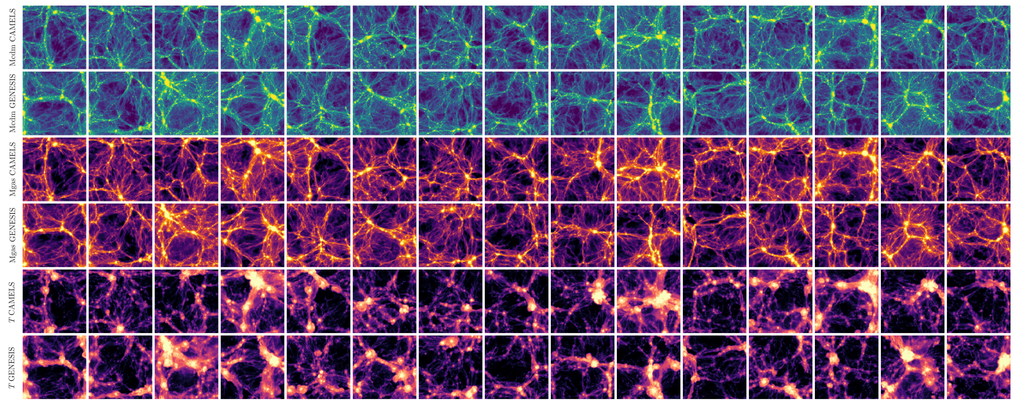

Figure 17. LH 0074: $(\Omega_{\rm m}, \sigma_8, A_{\rm SN1}, A_{\rm SN2}, A_{\rm AGN1}, A_{\rm AGN2}) = (0.282, 0.724, 1.587, 0.270, 1.330, 0.990)$. Each tile arranges 15 generated samples with the corresponding CAMELS reference rows.

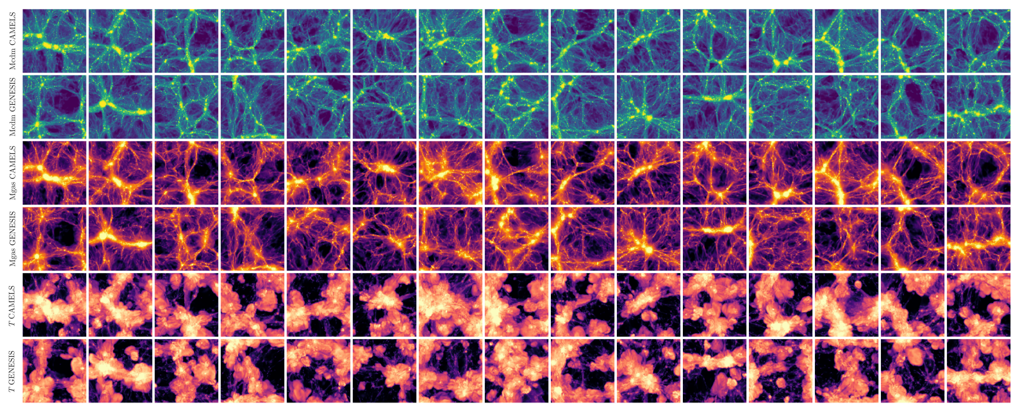

Figure 18. LH 0163: $(\Omega_{\rm m}, \sigma_8, A_{\rm SN1}, A_{\rm SN2}, A_{\rm AGN1}, A_{\rm AGN2}) = (0.337, 0.950, 0.577, 1.705, 0.670, 1.077)$. Each tile arranges 15 generated samples with the corresponding CAMELS reference rows.

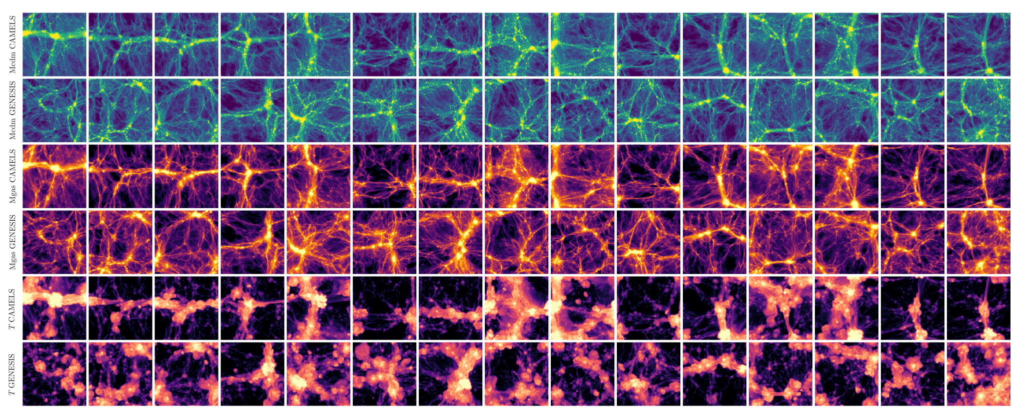

Figure 19. LH 0008: $(\Omega_{\rm m}, \sigma_8, A_{\rm SN1}, A_{\rm SN2}, A_{\rm AGN1}, A_{\rm AGN2}) = (0.389, 0.778, 0.287, 0.258, 1.699, 1.036)$. Each tile arranges 15 generated samples with the corresponding CAMELS reference rows.

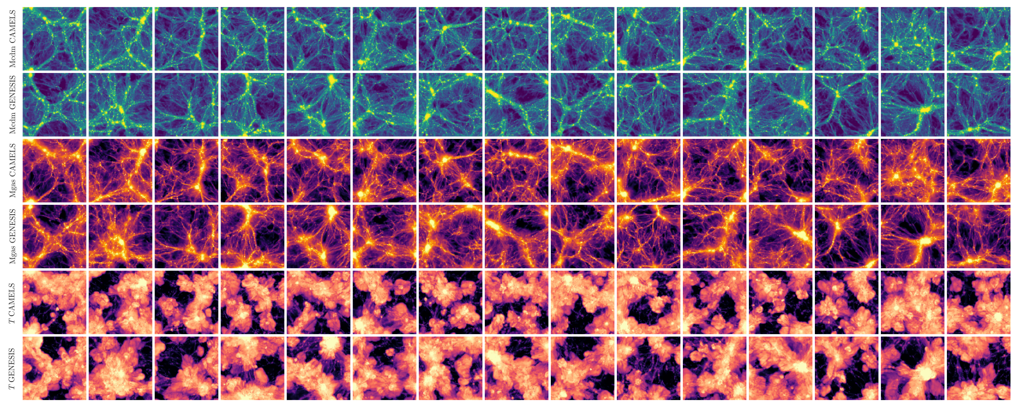

Figure 20. LH 0142: $(\Omega_{\rm m}, \sigma_8, A_{\rm SN1}, A_{\rm SN2}, A_{\rm AGN1}, A_{\rm AGN2}) = (0.425, 0.810, 0.274, 1.696, 0.621, 1.477)$. Each tile arranges 15 generated samples with the corresponding CAMELS reference rows.

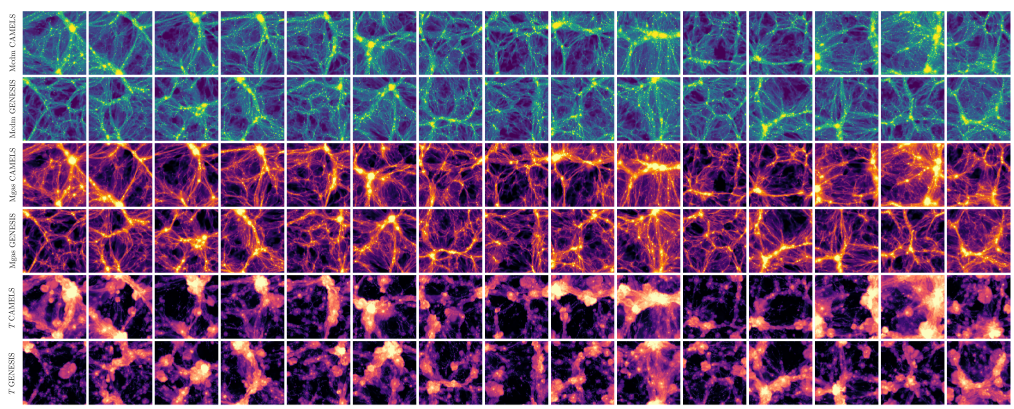

Figure 21. LH 0479: $(\Omega_{\rm m}, \sigma_8, A_{\rm SN1}, A_{\rm SN2}, A_{\rm AGN1}, A_{\rm AGN2}) = (0.462, 0.940, 0.993, 0.433, 1.762, 1.469)$. Each tile arranges 15 generated samples with the corresponding CAMELS reference rows.

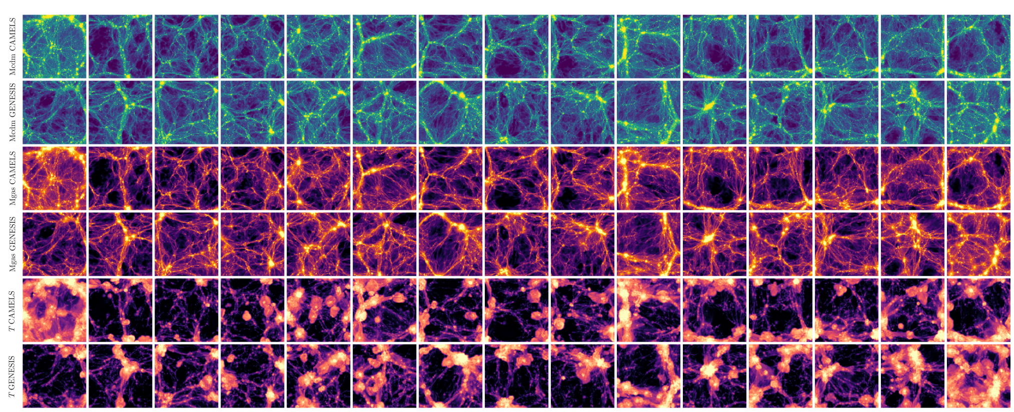

Figure 22. LH 0108: $(\Omega_{\rm m}, \sigma_8, A_{\rm SN1}, A_{\rm SN2}, A_{\rm AGN1}, A_{\rm AGN2}) = (0.499, 0.630, 1.310, 2.385, 0.518, 0.623)$. Each tile arranges 15 generated samples with the corresponding CAMELS reference rows.

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